



AI-Assisted Staff Attendance & Monthly Reporting System

System documentation: project report, user manual, database dictionary, API reference, installation guide, and test results for the system built to replace the manual monthly attendance-analysis process to the CAO.

PREPARED FOR	REPORTING PERIOD DEMONSTRATED	STACK	STATUS
Office of the CAO	June 2026	PHP · MySQL · Tailwind CSS	Working prototype

01 Project Report

Problem statement, objectives, AI methodology, and system architecture.

Introduction

Mbarara District Local Government's HR Unit maintains a daily staff attendance register: every staff member signs in before 8:00 a.m. and signs out after 5:00 p.m. At month end, HR is required to compile a Monthly Staff Attendance Analysis Report for the Chief Administrative Officer (CAO) — summarising attendance, punctuality, absenteeism, and flagging staff who need supervisory attention.

This process is currently manual: paper registers are tallied by hand, and the monthly narrative report is typed from scratch every time. This document describes an AI-enabled system built to replace that process — digitising the register, computing statistics automatically, flagging anomalies, forecasting next month's trend, and drafting the monthly report, while keeping HR in control of what is actually sent.

Problem Statement

1. **Manual tallying** — paper sign-in sheets counted by hand for every department, every month.
2. **No early warning** — chronic lateness or absenteeism only becomes visible once someone notices it, often long after the pattern is established.
3. **No forward view** — the CAO only sees a rear-view mirror of what already happened.
4. **Repetitive report writing** — the same kind of narrative memo is retyped from scratch every month.

Objectives

General objective: design and build an AI-enabled system that supports the daily register and automates the monthly analysis and reporting function for the CAO.

- Digitise sign-in/sign-out, enforcing the official 08:00 start and grace period automatically.
- Compute attendance, punctuality, and absenteeism rates per department and district-wide without manual tallying.
- Automatically flag staff who exceed policy thresholds for lateness or absence.

- Forecast next month's attendance and absenteeism trend from historical data.
- Auto-draft the narrative monthly report, while leaving HR the final decision to review and send it.
- Give the CAO a self-service, read-only view — no more waiting for a printed memo.

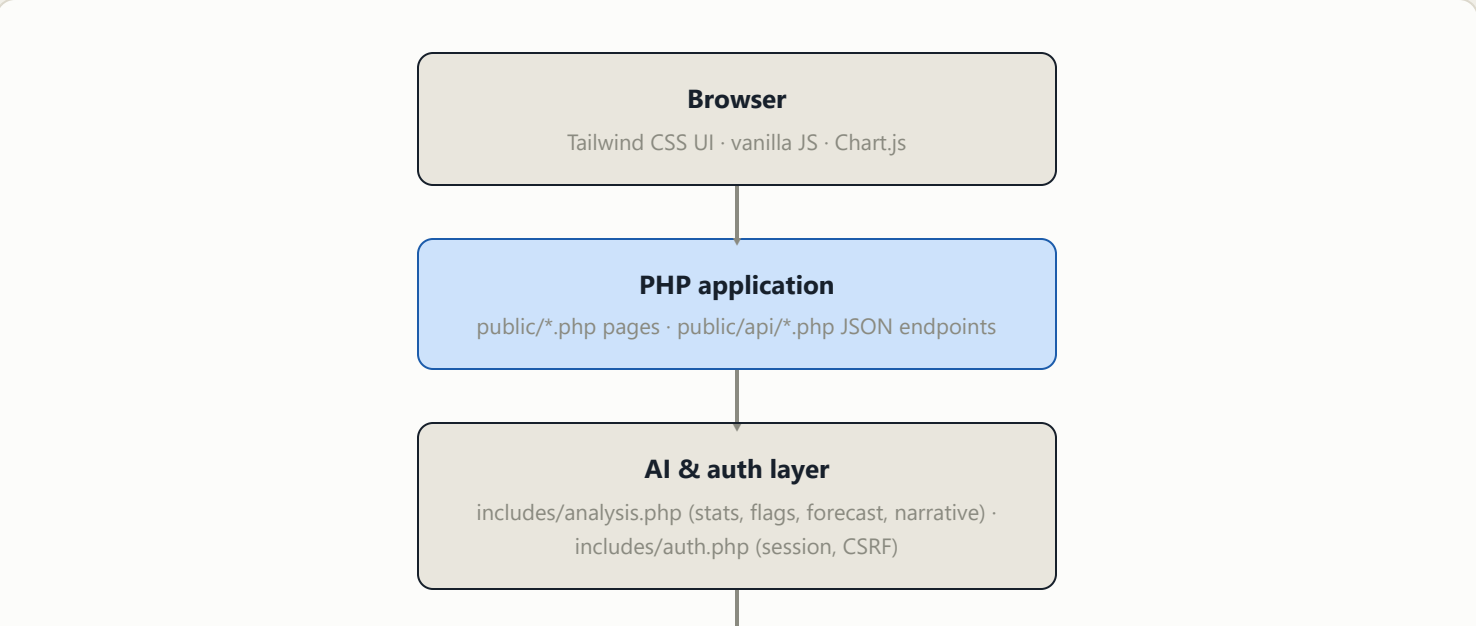
Why This Counts as "AI-Enabled"

Rather than a black-box model the HR unit couldn't explain to the CAO, the system uses three concrete, explainable AI techniques:

TECHNIQUE	WHAT IT DOES	WHY THIS APPROACH
Rule-based anomaly detection	Flags staff crossing configurable thresholds (default: >3 late arrivals or >2 unexplained absences/month)	Fully auditable — the CAO can see exactly why a name is flagged
Statistical forecasting	Fits a linear trend across trailing months to project next month's attendance/absenteeism, with an honest confidence label	Forward-looking signal without needing a training dataset the district doesn't have
Template-driven NLG	Turns computed statistics into the narrative prose HR currently types by hand	Drafts in seconds; HR still reviews before sending — human stays in the loop

`generate_ai_narrative()` is written as a deliberately swappable function — a richer, model-generated narrative (e.g. via the Claude API) is a drop-in replacement; the statistics, flags, and forecast stay as structured data either way.

System Architecture



MySQL — mbarara_attendance

staff · departments · users · attendance_records ·
monthly_reports · report_flags

Server-rendered PHP, no framework, no build step — easy to host on any shared server or XAMPP. Tailwind loads via CDN for this prototype.

Technology Stack

LAYER	CHOICE	RATIONALE
Backend	PHP 8 (mysqli, prepared statements)	No framework overhead; runs on any shared host/XAMPP; matches the assignment's required stack
Database	MySQL / MariaDB	Required by the assignment; relational model fits attendance/report structure
Frontend	Tailwind CSS (CDN) + vanilla JS	Fast to build, no build pipeline
Charts	Chart.js (CDN)	Lightweight, no extra backend dependency
Auth	PHP sessions + CSRF + password_hash()	Standard, framework-free security baseline

Limitations & Future Work

- The forecast uses a transparent linear trend rather than a deep model, since the district doesn't yet have years of digitised history — accuracy improves automatically as more months accumulate.
- No biometric hardware integration yet; the data model already supports it via capture_method .
- The narrative generator is template-based, with a documented seam for a future LLM-backed version.
- Tailwind is CDN-loaded for the prototype; production should compile a static bundle.

A worked example for **June 2026** is included as seeded demonstration data — 37 staff across 8 departments, with an 84.5% attendance rate and 14 staff flagged for policy-threshold review. See §6 for verification detail.

02 **User Manual**

Step-by-step guide for the HR Officer and CAO roles.

Signing In

ROLE	USERNAME	DEFAULT PASSWORD	CAN DO
HR Officer	hr.officer	password123	Everything below
CAO	cao	password123	Monthly Report only (read-only)

Change these passwords before real use — see §5, step 5.

Daily Register — recording attendance

The register has two dedicated panels rather than one shared form:

- **Check In** — lists only staff who have *not yet* checked in for the selected date. Once recorded, that person disappears from this list immediately.
- **Check Out** — lists only staff who checked in but have *not yet* checked out. They disappear from this list once recorded.

The system applies the attendance policy automatically: 08:00 start with a 15-minute grace period — anyone checking in after 08:15 is marked **Late** rather than **Present**. The Ledger below the form shows every entry for the date, filterable by department and status, with a Delete action per row for corrections.

Staff & Departments

Staff: Add, Deactivate (soft — keeps history), Reactivate, or Delete (hard — removes attendance history too, with confirmation). **Departments:** Add, rename inline, or delete — deletion is blocked while staff are still assigned, to prevent orphaned records.

Monthly Report — preparing and sending to the CAO

Opens on the most recently completed month by default. Shows summary tiles, department attendance bars, a daily trend chart, the AI insight panel (forecast + anomaly bullets), flagged-staff table, and the auto-drafted transmittal memo. Selecting **Send report to CAO** persists the report so it becomes visible on the CAO's own login, with a "Sent" banner and timestamp.

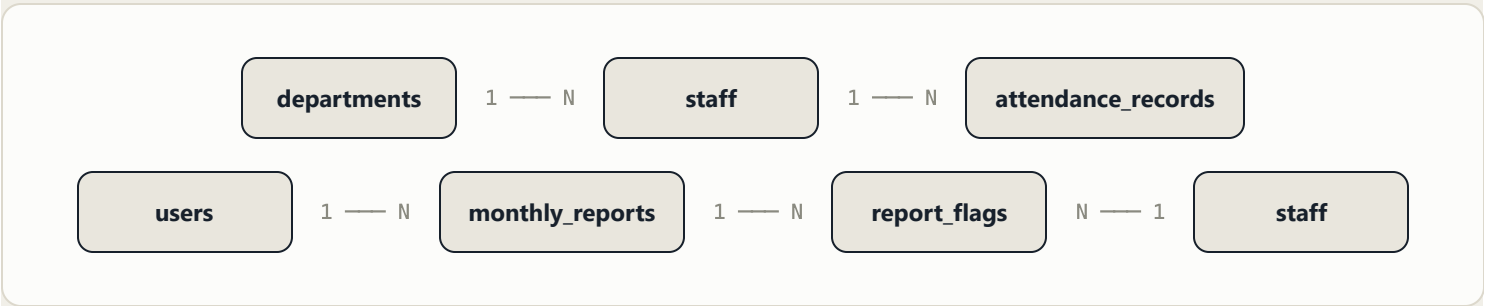
For the CAO

Sign in and land directly on Monthly Report — the only page available, and read-only. Use the period selector to review any month HR has sent, and **Print / Save PDF** for a hard copy.

03 Database Dictionary

Database `mbarara_attendance` — full DDL in `database/schema.sql`.

Entity Relationships



Core Tables

TABLE	PURPOSE	KEY CONSTRAINT
departments	The district's organisational units	UNIQUE(name)
staff	Establishment register — staff number, designation, department, status	UNIQUE(staff_no) ; delete cascades to attendance
users	Login accounts — hr / cao / admin roles	UNIQUE(username) ; bcrypt password hash
attendance_records	One row per staff member per day: check-in, check-out, status, remarks	UNIQUE(staff_id, attendance_date)
monthly_reports	Snapshot of each report sent to the CAO	UNIQUE(report_year, report_month)
report_flags	Which staff were flagged at the time a report was sent (audit trail)	FK → monthly_reports , staff
settings	Attendance policy as data — start time, grace period, thresholds, targets	PRIMARY KEY(setting_key)

Attendance Policy Defaults (settings)

KEY	DEFAULT	MEANING
day_start_time	08:00:00	Official reporting time
grace_minutes	15	Minutes before a check-in is marked Late
late_flag_threshold	3	Late arrivals/month before flagging
absence_flag_threshold	2	Unexplained absences/month before flagging
attendance_target_pct	85	District attendance-rate target

KEY	DEFAULT	MEANING
punctuality_target_pct	90	District punctuality-rate target

Full column-by-column reference (types, nullability, all FKs): docs/03-Database-Dictionary.md in the project repository.

04 API Reference

All endpoints under public/api/ return JSON and require an authenticated session; state-changing requests require a CSRF token.

ENDPOINT	METHOD	ROLES	PURPOSE
api/ledger.php	GET	hr, admin	Full attendance ledger for one date
api/staff_status.php	GET	hr, admin	Who still needs check-in vs. check-out for a date
api/record_entry.php	POST	hr, admin	Record a check-in or check-out
api/delete_entry.php	POST	hr, admin	Delete one attendance record
api/send_report.php	POST	hr, admin	Compute and persist the monthly report, visible to the CAO

Core AI Functions

Called server-side from dashboard.php and reports.php — not separate HTTP endpoints, but the system's core AI surface:

FUNCTION	PURPOSE
compute_period_stats()	Attendance/punctuality/absenteeism rates for a month
compute_department_stats()	Per-department breakdown, sorted best → worst
detect_flagged_staff()	Rule-based anomaly detection against policy thresholds
forecast_next_month()	Linear-regression forecast with a confidence label
generate_ai_narrative()	Template-driven NLG — the transmittal memo text

Full request/response payloads: docs/04-API-Reference.md in the project repository.

05 Installation Guide

Requirements: PHP 8.0+ with `mysqli` , MySQL 5.7+/MariaDB 10.3+, a web server.

Setup steps

1. Copy the project into `C:\xampp\htdocs\` (or point any web server's document root at its `public/` folder). Start Apache and MySQL.
2. Import the schema: `mysql -u root < database/schema.sql`
3. Seed a three-month demonstration history: `php database/seed_june2026.php`
4. To fill the current, still-open month without touching real entries: `php database/seed_fill_month.php <year> <month>`
5. Log in with the default accounts (below) — **then change both passwords** before real use.

Default accounts

ROLE	USERNAME	PASSWORD
HR Officer	<code>hr.officer</code>	<code>password123</code>
CAO	<code>cao</code>	<code>password123</code>

Production hardening checklist

- Change both default passwords via `password_hash()` — no in-app change-password screen yet.
- Serve over HTTPS; set `session.cookie_secure = 1` .
- Move database credentials to environment variables (`MBR_DB_HOST` , `MBR_DB_NAME` , `MBR_DB_USER` , `MBR_DB_PASS`).
- Review `settings` against the district's real attendance policy.

Full step-by-step guide including bulk CSV import guidance: `docs/05-Installation-Guide.md` .

06

Test Plan & Results

Testing was performed against a live MySQL instance, driving actual HTTP endpoints with authenticated sessions — not mocked.

Results summary

AREA	CASES	RESULT
Authentication & access control	4	All passed
Daily Register (check in/out, deletion, list scoping)	6	All passed
Staff & department management	8	All passed
Monthly report & AI layer	6	All passed

June 2026 verification figures: **84.5%** attendance rate, **85.5%** punctuality, **8.8%** absenteeism, across 37 staff and 22 working days — with 14 staff correctly flagged against policy thresholds.

Defects found and fixed during development

DEFECT	ROOT CAUSE	FIX
Flagging query threw a SQL exception	<code>HAVING</code> clause referenced aliases MySQL rejected alongside <code>OR</code>	Rewrote to repeat full <code>SUM()</code> expressions
Seed data showed 0% attendance	<code>bind_param()</code> type-string mismatch silently corrupted enum values	Corrected type string; verified with direct SQL
Two further type-string mismatches	Same class of bug, found by auditing every <code>bind_param</code> call	Corrected both; re-tested end to end
Unrealistic lateness in seed data	Baseline arrival jitter alone exceeded the grace period	Tightened the arrival-time model to match department bias
Trend chart grew taller on every redraw	Unsize canvas parent + <code>maintainAspectRatio: false</code> — a known Chart.js feedback loop	Wrapped canvas in a fixed-height container

Known gaps

- No automated regression suite yet — testing to date is manual/scripted, run per change.
- No department-reassignment UI for existing staff.
- No in-app password-change screen.

Full case-by-case results: `docs/06-Test-Plan-and-Results.md` .

07

Software Requirements Specification

Conforms to the structure of IEEE Std 830-1998, scoped to the system as designed and implemented — Version 1.0.

1 • Introduction

Purpose. This section specifies the functional and non-functional requirements of the system. It is the authoritative statement of what the system must do, to be read alongside §1 (rationale) and §3 (data model).

Intended audience. The HR Unit and CAO's office as operators; the assessor evaluating this work against the assignment brief; any future developer extending the system.

Product scope. Digitises the daily sign-in/out register and automates the monthly attendance-analysis reporting function HR currently performs by hand. Payroll, leave-application workflow, and biometric hardware integration are explicitly out of scope for this version.

2 • Overall Description

USER CLASSES

CLASS	DESCRIPTION
HR Officer	Operates the daily register, manages staff/department records, reviews and sends the monthly report
Chief Administrative Officer	Reviews sent monthly reports and flagged staff, read-only
System Administrator	Role reserved in the data model (<code>admin</code>) for future account/settings management, not yet exposed in the UI

CONSTRAINTS

- Must use PHP, JavaScript, and Tailwind CSS with a MySQL backend — the assignment-mandated stack.
- No client-side framework or build pipeline, so the system stays deployable on ordinary shared/government hosting.
- Biometric hardware and payroll integration are out of scope; the data model reserves a `capture_method` field for the former.

ASSUMPTIONS

- Each staff member belongs to exactly one department at a time.
- The working week is Monday–Friday; the 08:00 start time and 15-minute grace period are configurable but assumed uniform across departments.
- Forecasting requires at least one prior month of data; with none, the system reports "not available" rather than guessing.

3 · Functional Requirements

Each requirement carries a priority — **Must** (required for the brief), **Should** (materially improves the system), or **Could** (explicitly optional).

AUTHENTICATION & ACCESS CONTROL

ID	REQUIREMENT	PRIORITY
FR-1	The system shall require a valid username and password before granting access to any page other than login.	Must
FR-2	Passwords shall be stored only as bcrypt hashes, never in plain text.	Must
FR-3	Each page/endpoint shall be restricted to its authorised roles.	Must
FR-4	State-changing requests without a valid CSRF token shall be rejected.	Must

DAILY ATTENDANCE REGISTER

ID	REQUIREMENT	PRIORITY
FR-6	HR shall be able to record a check-in for any active staff member (time, method, remark).	Must
FR-7	HR shall be able to record a check-out for a staff member already checked in that date.	Must
FR-8	A check-in past the grace period shall be classified Late automatically; otherwise Present.	Must
FR-9	The Check-In list shall exclude staff who already checked in for the date.	Must
FR-10	The Check-Out list shall exclude staff who haven't checked in, or already checked out.	Must
FR-13	At most one attendance record shall exist per staff member per date.	Must

ESTABLISHMENT MANAGEMENT

ID	REQUIREMENT	PRIORITY
FR-14	HR shall be able to add a staff member with a unique staff number, name, designation, and department.	Must
FR-15	HR shall be able to deactivate a staff member without losing their history.	Must
FR-17	HR shall be able to permanently delete a staff member and their history, with confirmation.	Should

ID	REQUIREMENT	PRIORITY
FR-20	A department with active staff assigned shall not be deletable; the system shall state how many staff are blocking it.	Must

AI-ASSISTED ANALYSIS, FLAGGING & FORECASTING

ID	REQUIREMENT	PRIORITY
FR-21	The system shall compute district-wide attendance, punctuality, and absenteeism rates for any given month.	Must
FR-22	The same rates shall be computed per department, ranked highest to lowest.	Must
FR-25 / FR-26	Staff exceeding the configured late-arrival or absence threshold shall be flagged automatically.	Must
FR-29	Next month's attendance/absenteeism rate shall be projected via a linear trend across trailing months.	Must
FR-30	The forecast shall be labelled with a trend direction and confidence level, not presented as a bare number.	Must
FR-31	The system shall state plainly when there isn't enough data to forecast, rather than fabricating a figure.	Must
FR-32	The system shall auto-draft a narrative report incorporating the rates, best/worst department, flagged-staff count, and forecast.	Must

REPORT TRANSMITTAL & CAO REVIEW

ID	REQUIREMENT	PRIORITY
FR-33	A report shall only become visible to the CAO after an explicit HR "Send report to CAO" action — never auto-sent.	Must
FR-38	The CAO shall be able to view any previously sent report, including narrative, statistics, and flagged staff.	Must
FR-39	The CAO shall not be able to modify any record or send a report.	Must
FR-40	The system shall show when a viewed report was transmitted.	Should

4 • Non-Functional Requirements

ID	CATEGORY	REQUIREMENT
NFR-3	Security	All database access shall use parameterised/prepared statements — no string-concatenated SQL.
NFR-4	Security	All state-changing requests shall carry a CSRF token.

ID	CATEGORY	REQUIREMENT
NFR-7	Reliability	A unique constraint on (staff, date) shall guarantee one attendance record per person per day, even under concurrent requests.
NFR-9	Usability	Destructive actions shall require explicit confirmation before executing.
NFR-11	Usability	Error messages shall state what went wrong and what to do about it.
NFR-12	Portability	The system shall run unmodified on any standard LAMP/WAMP/XAMPP stack — no proprietary hosting features required.

5 · Traceability (excerpt)

REQUIREMENTS	IMPLEMENTED IN
FR-1 – FR-4	<code>includes/auth.php</code> , <code>public/index.php</code>
FR-6 – FR-13	<code>public/register.php</code> , <code>public/api/record_entry.php</code> , <code>public/api/staff_status.php</code>
FR-14 – FR-20	<code>public/staff.php</code> , <code>public/departments.php</code>
FR-21 – FR-32	<code>includes/analysis.php</code>
FR-33 – FR-40	<code>public/api/send_report.php</code> , <code>public/reports.php</code>

Full requirement set (40 functional, 15 non-functional requirements) and glossary:
`docs/07-Software-Requirements-Specification.md` .